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A Novel Lightweight CNN for Constrained IoT Devices: Achieving High Accuracy with Parameter Efficiency on the MSTAR Dataset

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ABSTRACT Convolutional neural networks (CNNs) have revolutionized fields such as image classification, natural language processing (NLP), and object detection. Remote sensing is no exception, particularly synthetic aperture radar (SAR) image analysis, where it addresses the challenge of classifying inherently noisy and indistinct images. Traditionally, this task relied heavily on manual intervention, making it time-intensive. With automation becoming increasingly critical, research has shifted toward deep learning techniques. However, many existing approaches are computationally intensive and require substantial memory resources. In this study, we present a lightweight CNN specifically designed for SAR applications, tested on the moving and stationary target acquisition and recognition (MSTAR) dataset. Our model surpasses previous studies by demonstrating higher accuracy while utilizing significantly fewer parameters. This novel architecture achieves an optimal balance between accuracy and computational efficiency. This is particularly useful in a resource-constrained environment in many real-world applications. More specifically, our proposed CNN model demonstrates robust performance across various scenarios, achieving an accuracy of 99.7% in classifying the three target classes under standard operating conditions (SOCs). Furthermore, when extended to classify ten classes, our proposed model outperforms several baseline algorithms from state-of-the-art literature. This research attempts to achieve a trade-off between model performance and model size, contributing to the development of CNNs suitable for resource-limited applications, particularly targeting IoT deployments. The findings present a practical solution for situations that prioritize both accuracy and resource conservation, thereby advancing discussions on efficient model design.

INDEX TERMS Automatic Target Recognition (ATR), CNN, Synthetic Aperture Radar (SAR), Image Classification, Internet of Things (IoT)

I. INTRODUCTION

SYNTHETIC aperture radar (SAR) is a revolutionary technology in remote sensing, capable of generating its own illumination, making it independent of external light sources such as the sun. This unique feature allows SAR to produce high-resolution images continuously, regardless of weather conditions, ensuring uninterrupted functionality around the clock [1]. SAR image target recognition is crucial for various critical applications, including enemy identification, battlefield surveillance, disaster relief, mineral exploration, oil spill detection, navigation, and military reconnaissance. Consequently, SAR automatic target recognition (ATR) presents a challenging yet vital problem that has been the focus of extensive research for many years [2]. The

primary goal of SAR ATR is to classify distinctive features extracted from SAR images into predefined categories using specific decision-making mechanisms [3]. However, the inherent imaging mechanism and speckle noise in SAR images pose unique challenges, distinguishing them from general optical images and complicating the task of target recognition.

Typically, the process of classifying SAR target images involves three main steps: preprocessing, feature extraction, and classification. In the domain of SAR image analysis, three commonly employed ATR paradigms are template matching, model-based techniques, and machine learning. Template matching, the most prevalent method, involves storing physical and structural features from training samples in a template dataset and comparing target features to determine matches.

Despite its popularity, this approach requires significant computational power and prior knowledge, with manual feature crafting being particularly challenging due to the complexity of data interconnections [4]. Model-based classification techniques replace target feature templates with scattering center models or solid models, constructing feature templates in real-time to aid recognition based on specific factors such as target posture. While advantageous, this method is complex and suitable only for specific situations [3].

Machine learning, particularly deep learning, has recently gained immense popularity due to its versatility and effectiveness in applications such as face detection, image recognition, and speech processing. However, deep learning techniques often encounter challenges related to computational complexity and memory requirements, especially when deployed on resource-constrained platforms like Internet of Things (IoT) devices [5]. For instance, the extensive array of parameters in convolutional neural networks (CNNs) poses a practical challenge in SAR image recognition, necessitating a focus on minimizing network parameters and computational resources, without compromising accuracy [6].

Given the limitations of existing ATR paradigms, there is a clear need for a more efficient and effective approach tailored specifically to the unique characteristics of SAR imagery in resource-constrained environments. Addressing these challenges, this paper proposes a novel lightweight CNN architecture specifically designed for SAR ATR tasks and rigorously tested on the moving and stationary target acquisition and recognition (MSTAR) dataset. Our objective is to develop a model capable of efficient inference without sacrificing classification accuracy, thereby facilitating real-time target recognition in SAR imagery. By leveraging the unique characteristics of SAR images and the requirements of ATR applications, we aim to create a compact yet effective CNN model suitable for deployment in resource-constrained IoT devices.

The study on the novel lightweight CNN for the MSTAR dataset makes several significant contributions to the field of SAR target recognition, summarized below:

- 1) The study introduces a novel, simple, lightweight CNN architecture optimized for SAR ATR tasks. This architecture reduces computational overhead without sacrificing accuracy, addressing the resource constraints commonly faced in real-world applications.
- 2) The proposed CNN achieves an optimal balance between accuracy and computational efficiency, surpassing previous studies by demonstrating higher accuracy while utilizing significantly fewer parameters. This balance is crucial for real-time target recognition in SAR imagery.
- 3) Detailed experiments are conducted to empirically analyze the proposed model under various parameter configurations. The study provides the model's performance under different settings, showcasing its robustness and effectiveness across various scenarios.

- 4) The study implements training optimizations such as adjusting the learning rate, utilizing validation data, and visualizing the training progress. These strategies contribute to fine-tuning the CNN model, resulting in a well-performing and generalizable model for SAR target recognition tasks.
- 5) Addressing the unique characteristics of SAR images and the requirements of ATR applications, the study aims to create a compact yet effective CNN model suitable for deployment in resource-constrained IoT devices. This approach fills a gap in existing ATR paradigms, providing a more efficient and effective solution for SAR target recognition.

The remainder of this paper is structured as follows: Section 2 provides an overview of related work in SAR ATR and discusses the challenges associated with traditional CNN architectures. Section 3 presents the architecture of our proposed lightweight CNN model, elucidating its design rationale and training procedures. Section 4 offers an overview of the dataset and model evaluation. Section 5 details the experimental results and discussions, including performance evaluation metrics. Section 6 outlines potential future work. Finally, Section 7 concludes the study by summarizing key findings and outlining avenues for further research.

II. RELATED WORK

The field of SAR target recognition has made substantial progress in recent years, owing to the increased availability of SAR imagery and the increasing need for automated target detection and classification algorithms. In this section, we review the existing literature on SAR target recognition techniques, with a particular focus on approaches developed to address the challenges of accuracy, computational efficiency, and parameter optimization, especially in the context of the limited data available in datasets such as MSTAR.

One of the techniques extensively used in the literature is template matching. This method involves matching extracted features with different template classes. In their study, [3] utilized the binary target region. They further improved upon these differences by applying the Euclidean distance transform to the region residuals. [7] suggested a SAR ATR approach that reconstructs the target using attributed scattering centers (ASCs). They used a neighbor matching technique to determine correspondence between the template set and the test set of ASCs. [8] proposed a target identification technique for SAR that involves matching ASCs to binary target areas. [9] developed a statistics-based distance measurement method to determine the distance between individual ASCs. Then they used the Hungarian technique to build a one-to-one correlation between two sets of ASCs. [10] used an adaptive Gaussian distribution to characterize the uncertainty related with individual ASC extraction based on characteristics. Subsequently, they defined the distance between two distinct ASCs as the Kullback-Leibler (KL) divergence between the Gaussian distributions that represent the uncertainty of the two ASCs. A cost matrix was then cre-

ated using the proposed distance measure, which accounts for the costs associated with missing and false ASCs, establishing a one-to-one connection between the two sets of ASCs. [11] introduced a method for matching scattering centers through sequential feature matching. They derived features from point sets obtained from scattering centers and estimated them using the ASCs model. However, due to noise or occlusion, there are often missing or incorrect scattering centers in the extracted or evaluated set. Errors in extraction can lead to difficulties in matching, compounded by the fact that SAR images are sensitive to target pose, rendering scattering centers obtained from an image valid only within nearby angles [10]. Additionally, the lack of suitable templates significantly impacts performance under extended operating conditions (EOCs) [12]. Furthermore, the sophisticated scattering center model makes retrieval of scattering centers more challenging compared to normal geometrical properties such as shadow and target region [3]. The presence of missing alarms (MAs) or false alarms (FAs) results in an unordered scattering center list with varying numbers of scattering centers, making it nearly impossible to directly train classifiers like sparse representation-based classification (SRC) or support vector machines (SVM) using scattering centers [9].

Ensemble classifiers and feature fusion techniques are also utilized in SAR ATR, combining multiple approaches to enhance efficiency. [13] introduced a hierarchical fusion technique for integrating complementary features in SAR ATR. They employed three features: ASCs, target outline, and principal component analysis (PCA). [14] introduced a method that relies on fusion of complementary features: local binary pattern (LBP), elliptical Fourier descriptors (EFDs), and PCA features are employed to characterize SAR images from different perspectives, aiming to provide more comprehensive representations of SAR targets. However, the elliptical Fourier series (EFS) method, which forms the basis of EFDs, relies on handcrafted features and is susceptible to issues such as target translation and invariance under speckle variation [15]. [16] introduced a hierarchical propelled strategy for multi-classifier fusion (HPSMCF), which integrates characteristics from both parallel and series structures. For feature extraction, they employed three techniques: PCA, non-negative matrix factorization (NMF), and linear discriminant analysis (LDA). However, LDA and PCA are susceptible to rotation and scale variations [17]. Additionally, PCA and LDA features represent the global attributes of targets and may lack clear physical meanings. Consequently, they may struggle to handle partial absences or occlusions of objects [9]. [18] introduced a novel framework for feature fusion based on CNN and scattering center features, marking the first instance of such integration. [19] presented an improved CNN model for addressing SAR ATR by augmenting features and employing ensemble learning techniques. [20] proposed SAR ATR through a hierarchical combination of ASC matching and CNN classification techniques. [21] introduced a dissimilar CNN ensemble, constructing the network with several filter stacks and a non-complete connection system to reduce

independent parameters and enhance training efficiency. [17] utilized pseudo-Zernike moments (PZM) for their rotation, scale, and translation invariant properties in feature extraction. For feature extraction, regions of interest (ROIs) in the form of texture, area, and boundary are utilized. However, geometry-based features like EFS and Zernike moments (ZMs) can obscure SAR target outlines, posing challenges in accurate detail extraction [9].

The researchers have employed various traditional machine learning techniques to address this problem. [22] utilized SVM for classification and the radial method for feature extraction. applied SVM for classification, while employing Flusser moments to characterize various properties of the target and its shadow, including surface estimation, and the relative position between the shadow and the target. However, these techniques necessitate hand-designed features and heavily rely on subjective elements, thereby limiting their degree of generality.

In recent years, a multitude of variations and combinations of CNNs have emerged as prominent tools for SAR ATR. [23] proposed an unsupervised deep learning model based on an encoding-decoding architecture. [24] developed a CNN-based SAR ATR technique utilizing the ASCs target reconstruction method. [4] introduced a super-resolution generative adversarial network (SRGAN) and deep convolutional neural network (DCNN) based SAR ATR approach. [25] devised a speckle-noise-invariant CNN employing regularization to mitigate feature variations caused by noise. [26] presented the all-convolutional networks (A-ConvNets), which utilizes sparsely linked layers instead of fully connected layers. However, several challenges persist in SAR object classification tasks utilizing CNNs. Insufficient labeled training data remains a common issue [23], while speckle noise continues to affect extracted features [25]. As the number of layers increases in CNNs, more parameters need to be trained, leading to potential overfitting and convergence issues [4]. Additionally, deploying original DCNNs on real-time embedded devices for SAR target recognition poses challenges due to their high computational cost and large memory overhead [27].

[2] devised SARNet, a lightweight CNN approach for classification, which utilizes a two-layer convolutional neural network and a clock-wise data augmentation technique to mitigate overfitting by generating sufficient SAR images. [6] proposed a lightweight CNN model for SAR image target recognition. It enhances feature extraction with visual attention mechanisms and reduces computational costs using depthwise separable convolution. Additionally, a new weighted distance measure loss function addresses data imbalance for improved recognition accuracy. [27] developed a lossless lightweight CNN design strategy for efficient SAR target recognition, utilizing pruning and knowledge distillation. Structured pruning is applied layer by layer to create a lightweight network, serving as the student network, which is then refined through knowledge distillation with the teacher network to restore accuracy. [28] introduces a lightweight

two-stream CNN. It initially extracts low-level features and then utilizes two streams to extract global and local features. The first stream uses global maximum pooling for local features, while the second uses large-stride convolution kernels for global features. Lastly, the streams are merged to enable target recognition.

Despite advancements in SAR target recognition techniques, several challenges persist, including speckle noise, limited training data, computational complexity, and robustness to variations in target appearance. Moreover, SAR images often present inherent complexities such as layover and shadow effects, further complicating recognition tasks.

A comparative analysis of different SAR target recognition approaches reveals trade-offs between computational efficiency, accuracy, and robustness. While template matching offers simplicity and speed, it may lack robustness in complex scenes. Model-based techniques provide flexibility but require accurate target models. Ensemble and fusion methods enhance accuracy but also elevate system complexity, resulting in increased computational demands and prolonged processing times. Machine learning methods offer automatic feature extraction but may suffer from overfitting and high computational costs.

In this study, we propose a novel lightweight CNN architecture to enhance SAR ATR classification performance for ground military vehicles and to reduce overfitting and computational costs.

III. METHODOLOGY

In this section, we introduce a lightweight CNN model, denoted as LSARNet, designed specifically for SAR ATR. In the following sub-sections, we will outline the architecture, implementation, and key features of the proposed network. Subsequently, we will delve into the specifics of model training.

A. ARCHITECTURE OF THE PROPOSED NETWORK

In addressing the imperative need for efficient and lightweight solutions in SAR target recognition, our proposed methodology aims to streamline the process by introducing a compact yet powerful two-layer convolution layer neural network. Illustrated in Figure 1, our LSARNet design optimally balances model complexity and performance, offering a promising solution for reducing parameters and enhancing efficiency.

The network architecture comprises two convolutional layers, labeled as Conv1 and Conv2, each followed by batch normalization and rectified linear unit (ReLU) activation layers. Conv1 generates 8 feature maps, while Conv2 produces 16 feature maps. The model consists of two pooling layers, Avgpool2d1 and Avgpool2d2, utilize average pooling. The classification stage involves a fully connected layer (FC) with 3 neurons, followed by a softmax classifier for finalizing the classification outcomes.

The architectural design gains advantages from incorporating batch normalization following each convolutional layer. This technique aids in stabilizing and expediting training

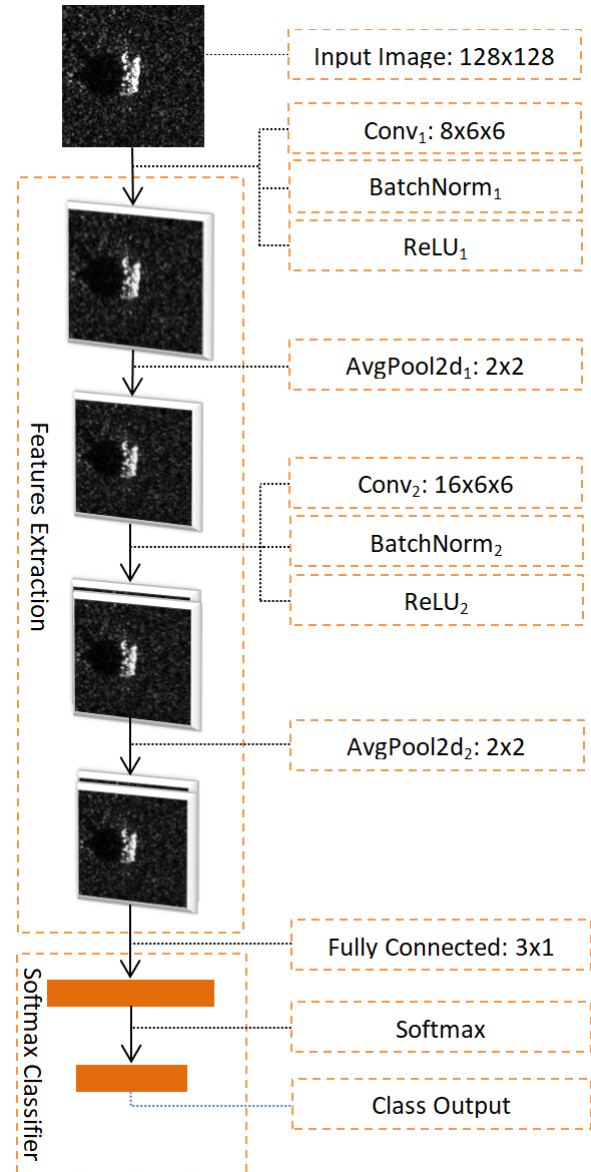


FIGURE 1. Operational Framework

processes by standardizing the input to subsequent layers, thereby addressing concerns associated with internal covariate shift. It augments the network's ability to generalize, leading to enhanced performance and smoother convergence during training iterations. Additionally, the inclusion of a ReLU activation layer after each batch normalization layer serves to reduce parameter interdependence and alleviate the risk of overfitting.

In our network design, we opted for average pooling over the commonly used max pooling, primarily due to the high noise levels inherent in SAR data. Pooling layers play a crucial role in downsampling feature maps, aiding in spatial reduction and capturing essential information.

While max pooling preserves the highest activation within a local area, average pooling calculates the mean activation.

Max pooling extracts the most active features within a region, emphasizing precise spatial localization and being sensitive to outliers. On the other hand, average pooling computes the average value in a region, providing a more generalized representation with less sensitivity to outliers.

The choice of average pooling is motivated by the need for a more robust and stable feature extraction process. We used average pool with stride 1 to mitigate sensitivity to outliers and to not lose critical information. As larger strides reduce structure and computational cost however it also loses important information due to which the model decreases accuracy as shown in Table 9. Furthermore, average pooling helps reduce overfitting by providing a more generalized representation of features. This can be beneficial when the training data is limited like in the case of MSTAR dataset. Notably, in experiments conducted on the MSTAR dataset, average pooling has demonstrated the capability to capture better features compared to max-pooling method as shown in Table 8.

In the output layer of the neural network, we utilized the softmax activation function to derive class probabilities, while cross-entropy loss was employed to define the objective function during training.

The softmax function is defined as follows for the raw scores

$$Z = (Z_1, Z_2, \dots, Z_n) \quad (1)$$

$$\text{softmax}(Z)_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}} \quad (2)$$

where e is the base of the natural logarithm. This function ensures that the output values are normalized probabilities, summing to 1 across all classes.

Additionally, the cross-entropy loss for a single training example with true class label y_i and predicted probabilities $\hat{y}_i = (\hat{y}_{i1}, \hat{y}_{i2}, \dots, \hat{y}_{ik})$ is calculated as:

$$\text{CrossEntropy}(y_i, \hat{y}_i) = - \sum_{j=1}^k y_{ij} \log(\text{Softmax}(z)_j) \quad (3)$$

This loss function measures the disparity between the actual distribution and the predicted probability distribution, guiding the model's learning process.

For a dataset with N training examples, the overall cross-entropy loss is calculated as the average over all examples:

$$\text{CrossEntropy}(\{y_i\}, \{\hat{y}_i\}) = \frac{1}{N} \sum_{i=1}^N \text{CrossEntropy}(y_i, \hat{y}_i) \quad (4)$$

The parameters are presented in Table 1 for the proposed model.

B. TRAINING DETAILS

The CNN requires a substantial amount of data to effectively train the model parameters. In this study, we utilized the MSTAR dataset, which contains a limited number of images

per class. To address this limitation and enhance the performance of the proposed method, we initially augmented the training set.

In the context of data augmentation, a single technique was employed to enhance the diversity of the training dataset. Rotation augmentation was applied by randomly rotating input images within a range of -10 to 10 degrees. This was achieved by employing a rotation matrix R_θ , where θ represents the random rotation angle. Mathematically, the rotation transformation applied to an image I can be represented as:

$$I_{\text{rotated}} = R_\theta(I) \quad (5)$$

where R_θ is the rotation matrix:

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (6)$$

This augmentation technique contributed to a more diverse and robust training dataset, facilitating improved model generalization.

• Hyperparameter Tuning Process

Hyperparameter tuning was conducted using manual search techniques. The process involved adjusting kernel sizes and the number of kernels, along with other training parameters, and analyzing the results through training progress plots. This iterative process allowed us to fine-tune the model and identify the optimal configuration for achieving high accuracy.

• Training Hyperparameters

In the training phase of our CNN, we employed a customized set of training options to optimize the model's performance and convergence. The optimization algorithm chosen for this task is stochastic gradient descent with momentum (SGD), which includes a momentum parameter set to 0.5 to accelerate convergence. The training process was limited to a maximum of 10 epochs, ensuring a balance between training time and model refinement. An initial learning rate was set to 0.001, and a piecewise learning rate schedule was adopted to dynamically adjust the learning rate during training. Specifically, the learning rate was decreased by a factor of 0.1 after 8 epochs, contributing to more refined weight updates. To monitor the model's progress and prevent overfitting, validation data were utilized, with validation checks performed every 5 epochs. The training process was visualized using a training progress plot. Additionally, mini-batches of size 16 were used to efficiently process the training data. This comprehensive set of training options was instrumental in fine-tuning our CNN, resulting in a well-performing and generalizable model for the given task.

IV. MSTAR DATASET AND MODEL EVALUATION

A. DATASET DESCRIPTION

In this study, the MSTAR dataset has been used, which is an openly available dataset consisting of ten different types of land vehicles. These ten ground military targets include D7, BRDM2, T62, 2S1, BTR70, T72, ZSU234, ZIL131, BTR60,

TABLE 1. LSARNet parameters configuration

Layers	Output size	Stride / Filter size	Learnables	Parameter
Input	$128 \times 128 \times 1$	-	-	-
Conv ₁	$128 \times 128 \times 8$	$1/6 \times 6$	Weights $6 \times 6 \times 1 \times 8$ Bias $1 \times 1 \times 8$	296
BatchNorm ₁	$128 \times 128 \times 8$	-	Offset $1 \times 1 \times 8$ Scale $1 \times 1 \times 8$	16
ReLU ₁	$128 \times 128 \times 8$	-	-	-
AvgPool2d ₁	$127 \times 127 \times 8$	$1/2 \times 2$	-	-
Conv ₂	$127 \times 127 \times 16$	$1/6 \times 6$	Weights $6 \times 6 \times 8 \times 16$ Bias $1 \times 1 \times 16$	4624
BatchNorm ₂	$127 \times 127 \times 16$	-	Offset $1 \times 1 \times 16$ Scale $1 \times 1 \times 16$	32
ReLU ₂	$127 \times 127 \times 16$	-	-	-
AvgPool2d ₂	$126 \times 126 \times 16$	$1/2 \times 2$	-	-
Fc	$1 \times 1 \times 3$	-	Weights 3×254016 Bias 3×1	762051

and BMP2. The MSTAR dataset was obtained by Sandia's Xband (9.6 GHz) HH-polarization spotlight SAR in 1995 and 1996, separately. These images are approximately 128×128 pixels with a resolution of $0.3 \text{ m} \times 0.3 \text{ m}$. They cover azimuth angles from 0 to 360 degrees and pitching angles of 15, 17, 30, and 45 degrees. The dataset is accessible through the sensor and data management system (SDMS) of the USA Air Force. A substantial body of research leveraging the MSTAR dataset is available in existing literature. Scholars utilize this dataset to evaluate their methods and compare them with previously published techniques [1]. Figure 2 depicts both optical and SAR images.

To assess the robustness and effectiveness of the proposed approach, we performed experiments under standard operating conditions (SOCs) using two widely recognized subsets of the MSTAR dataset, namely SOC-3 and SOC-10, which are extensively employed in the literature. SOC means that targets in a test set and training set have the same target configuration and serial number but different azimuths and slightly different depression angles.

1) SOC Datasets

For SOC, we considered both SOC-3 and SOC-10 datasets. SOC-3 contains three targets, i.e., T72, BTR70, and BMP2 whereas SOC-10 contains all categories of the MSTAR dataset. It is a perfect SOC under this scenario since there is only a 2° depression angle difference between the test and training samples. Images captured at a depression angle of 17° are utilized for training purposes, while those taken at a depression angle of 15° are employed for testing. The training and test sets under SOC-3 and SOC-10 are described in Table 2 and Table 3 respectively.

B. ITERATION OF TRAINING

During the training phase, we chose to fine-tune the models over 10 epochs while also documenting the output classification results and the loss values. As can be seen in Figure 3, the accuracy has improved notably during the initial stages of training and remains consistent thereafter. Limiting the

epochs to ten resulted in a considerable reduction in training time. The progression of the training process is visualized in Figure 3, showcasing the efficiency and robustness of the proposed network. Notably, the network exhibits accelerated convergence speed and maintains a relatively stable state at the end. For all the datasets used in this study, we randomly selected 80% for training and 20% for validation.

The experiments in this study were conducted using a Dell Latitude 7480 featuring an Intel Core i7 7th Generation processor, specifically the Intel(R) Core(TM) i7-7600U CPU @ 2.80GHz with four CPUs running at a base clock speed of approximately 2.9 GHz. The operating system employed was Windows 10 Pro 64-bit, and the hardware configuration included 16 GB of RAM and Intel(R) HD Graphics 620. The computational aspects of the research were facilitated by the use of MATLAB R2019a as the programming environment on this hardware setup.

C. EVALUATION OF THE PROPOSED LSARNET

This section examines major components of the LSARNet to better understand its lightweight architecture. The feature extraction accuracy of the network under various parameters is evaluated, and a complete examination of the impact of various parameters on test outcomes is explored in depth, spanning Tables 4 through 9.

In Table 4, we examined the effect of varying convolutional kernel sizes on the model's performance. Our experiments covered a range of kernel sizes from 3×3 to 10×10 for both Conv1 and Conv2 layers. The results reveal a notable trend: while smaller kernel sizes (e.g., 3×3) tend to capture fine-grained details, they may not adequately capture larger contextual features, which are crucial for tasks involving complex image data like SAR images. Conversely, larger kernels (e.g., 10×10) can capture more extensive features but may also lead to overfitting due to their larger parameter space. The choice of kernel size should be considered in relation to the size of the features in the data, as excessively large kernels can result in a loss of detail. Interestingly, our findings indicate that the model with a 6×6 kernel size in Conv1 and Conv2



FIGURE 2. Military targets: (above) optical images versus (below) SAR images [24].

TABLE 2. Number of train and test samples for SOC-3 dataset

Target Class	Training set		Test set	
	Depression	No of Images	Depression	No of Images
BMP2	17°	233	15°	195
BTR70	17°	233	15°	196
T72	17°	232	15°	196

achieved the highest accuracy of 99.7%. This optimal kernel size appears to strike a balance between capturing sufficient spatial information and avoiding overfitting. While odd-sized kernels are more conventionally used due to their ability to capture neighboring pixels symmetrically, our experiments revealed that the even-sized 6x6 kernel provided better results for this task. The broader receptive field of the 6x6 kernel allowed the model to capture more complex patterns in the SAR data, and its performance on the MSTAR dataset surpassed that of odd-sized kernels. Thus, although unconventional, the even 6x6 kernel was the most effective choice in our architecture based on empirical results. Kernels larger than 6x6 did not significantly improve performance and, in some cases, led to decreased accuracy. This suggests that while larger kernels can aggregate more information, they may also dilute the specificity required for effective feature extraction in SAR images.

As detailed in Table 5, we assessed how varying the number of convolutional layers impacts model performance. Our experiments compared configurations with two, three, and four convolutional layers. The results highlight that the model with two convolutional layers outperformed those with three or four layers, achieving an accuracy of up to 99.7%. This trend

underscores the principle that simply increasing the number of layers does not necessarily translate to better performance. More layers add complexity to the model, which can lead to increased computational costs and a higher risk of overfitting without corresponding gains in accuracy. Our results suggest that a well-structured model with fewer convolutional layers can be more effective, provided that each layer is optimized for extracting relevant features. The findings also imply that the model's architecture should be carefully designed to balance depth and complexity to achieve optimal performance.

Table 6 presents the effect of batch normalization layers on model accuracy. Model 4, utilizing these layers, achieved the highest accuracy, while model 1, without these layers, achieved the lowest accuracy of 75.8%. In the absence of batch normalization layers, model training stability is compromised, resulting in erratic learning and accuracy stagnation due to issues such as vanishing or exploding gradients and sensitivity to weight initialization. Batch normalization is crucial for stabilizing training, allowing for faster convergence, improved generalization, and enhanced performance by addressing these challenges. The inclusion of just one batch normalization layer after Conv1 or Conv2 led to a significant improvement in accuracy, reaching levels of up to

TABLE 3. Number of train and test samples for SOC-10 dataset

Target Class	Training set		Test set	
	Depression	No of Images	Depression	No of Images
BMP2	17°	233	15°	195
BTR70	17°	233	15°	196
T72	17°	232	15°	196
T62	17°	299	15°	273
BRDM2	17°	298	15°	274
BTR60	17°	256	15°	195
ZSU234	17°	299	15°	274
D7	17°	299	15°	274
ZIL131	17°	299	15°	274
2S1	17°	299	15°	274

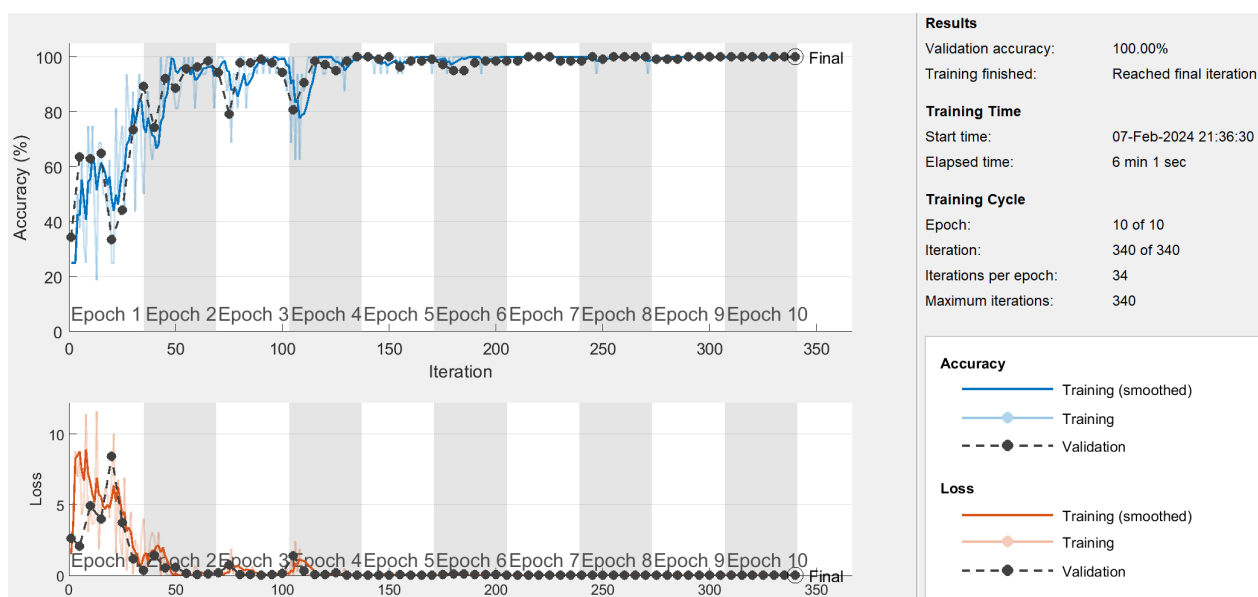


FIGURE 3. Training Progress

TABLE 4. Impact of varying convolutional kernel sizes

Model	Conv1	Conv2	Accuracy(%)
1	3/8	3/16	97.6
2	4/8	4/16	98.1
3	5/8	5/16	98.1
4	6/8	6/16	99.7
5	7/8	7/16	99.3
6	8/8	8/16	98.1
7	9/8	9/16	98.8
8	10/8	10/16	98.5

99%.

Table 7 presents the impact of ReLU layers on model accuracy. Model 4, utilizing ReLU layers, achieved the highest accuracy. In contrast, model 1, without ReLU layers, achieved the lowest accuracy. These findings emphasize the significant role of ReLU layers in enhancing model performance.

Table 8 demonstrates the influence of various pooling layers on model accuracy. Among the tested models, model 2

TABLE 5. Impact of convolutional layers

Model	Conv1	Conv2	Conv3	Conv4	Accuracy(%)
1	6/8	6/16	-	-	99.7
2	6/8	6/16	6/32	-	99.3
3	6/8	6/16	6/32	6/64	98.8

TABLE 6. Effect of batch normalization layers on the model

Model	Conv1	Batch Norm1	Conv2	Batch Norm2	Accuracy (%)
1	6/8	No	6/16	No	75.8
2	6/8	Yes	6/16	No	99.0
3	6/8	No	6/16	Yes	99.0
4	6/8	Yes	6/16	Yes	99.7

stands out, employing average pooling in both pooling1 and pooling2 layers, resulting in the highest accuracy achieved. These results underscore the significance of utilizing average pooling methods to enhance model performance.

TABLE 7. Effect of ReLU layer on the model

Model	ReLU1	ReLU2	Accuracy(%)
1	No	No	94.0
2	Yes	No	97.4
3	No	Yes	98.5
4	Yes	Yes	99.7

TABLE 8. Effect of pooling layers

Model	Pooling1	Pooling2	Accuracy(%)
1	Max	Max	98.6
2	Average	Average	99.7
3	Max	Average	98.6
4	Average	Max	97.8
5	No-Pooling	No-Pooling	98.0

Table 9 illustrates the impact of different pooling layers strides on model accuracy. Model 1, with strides of 1 in both pooling layers achieved high accuracy, indicating the importance of preserving spatial information at a finer resolution. However, increasing stride values in the pooling layers can lead to a decline in accuracy, as observed in rest of the models. Larger strides introduce challenges such as information loss and reduced feature resolution, which can hinder the network's ability to effectively capture patterns and details in the input data.

TABLE 9. Effect of strides in pooling layers

Model	Pooling1 stride	Pooling2 stride	Accuracy(%)
1	1	1	99.7
2	1	2	99.1
3	2	1	98.6
4	2	2	97.1
5	3	3	88.1
6	4	4	83.3

To gain further insight and showcase the effectiveness of the presented model, we visually analyze various layers in Figure 4. The proposed LSARNet demonstrates that reducing the number of filters in a model yields numerous benefits, as evidenced by the experiments. It decreases computation during training and inference, resulting in faster processing and lower costs. Additionally, it reduces the model's memory footprint, making it suitable for resource-constrained IoT devices. Moreover, fewer filters encourage the network to learn more essential features relevant to the target, improving generalization and reducing overfitting risks.

D. HEATMAP ANALYSIS OF THE PROPOSED LSARNET

The heatmap visualizations of various layers in our lightweight CNN reveal a progressive enhancement in the network's focus on the T72 tank target, demonstrating effective feature extraction and refinement. In Figure 5, which illustrates the conv1 layer, and Figure 7 for the conv2 layer,

regions of high activation, indicated by intense red hues, are observed not only on the target itself but also in areas close to it, including shadowed regions and other parts of the image. This distribution implies that these layers are responsive to a broad spectrum of features pertinent to the tank as well as other image elements. Additionally, regions marked by blue and light blue indicate areas of lower activation. Following the application of the ReLU activation function, as shown in Figures 6 and 8, the heatmaps reveal that high activation areas are now highlighted in red, yellow, and green, becoming more concentrated on the target. Conversely, dark blue and blue areas represent reduced activation levels in the remaining regions of the image. This shift underscores the network's ability to refine and concentrate its focus more accurately on the target after the non-linear transformation, thereby enhancing its overall classification efficacy. These findings align with existing literature on CNN interpretability, such as [29] and [30], which also emphasize the importance of understanding model focus in complex tasks.

E. COMPUTATIONAL COMPLEXITY

In the context of deep learning, understanding the computational complexity of a model is crucial for evaluating its computational efficiency and scalability. The computational complexity of our CNN architecture, often measured in floating-point operations (FLOPs), is primarily influenced by the convolutional and fully connected layers. The computational complexity for convolutional layers is calculated using the following equation:

$$2 \times (H_{in} \times W_{in} \times F^2 \times C_{in} \times C_{out}) \quad (7)$$

where H_{in} and W_{in} are the height and width of the input feature map, F is the filter size, C_{in} is the number of input channels, and C_{out} is the number of output channels. For batch normalization layers, the complexity is given by

$$2 \times H_{in} \times W_{in} \times C \quad (8)$$

where C is the number of channels. The average pooling layer's complexity is calculated as

$$4 \times (H_{out} \times W_{out} \times C) \quad (9)$$

For fully connected layers, the complexity is represented by

$$2 \times \text{input_units} \times \text{output_units} \quad (10)$$

accounting for both multiplication and addition operations. In our CNN, the FLOPs for the main layers are as follows:

- Convolutional Layers

Conv1 and Conv2 contribute approximately 0.0094 GFLOPs and 0.1486 GFLOPs, respectively.

- Fully Connected Layer

The FC layer contributes about 0.0015 GFLOPs.

The total FLOPs for the entire model is approximately 0.1627 GFLOPs for SOC3 and about 0.1678 GFLOPs for

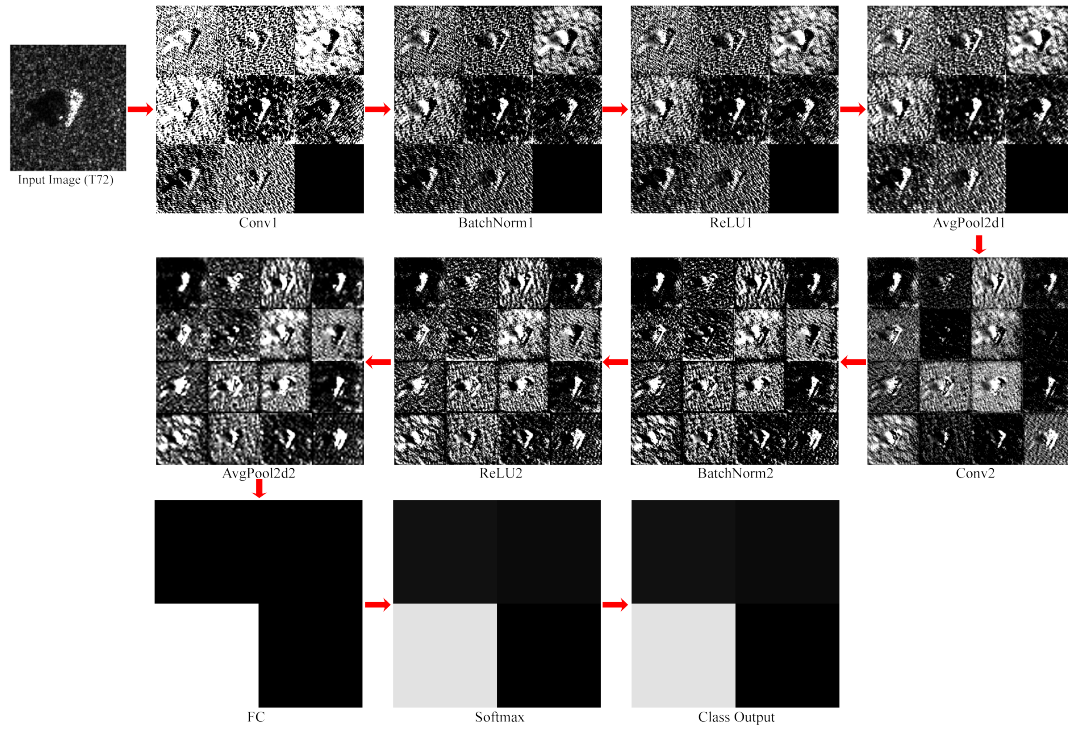


FIGURE 4. SAR image feature map visualization of T72

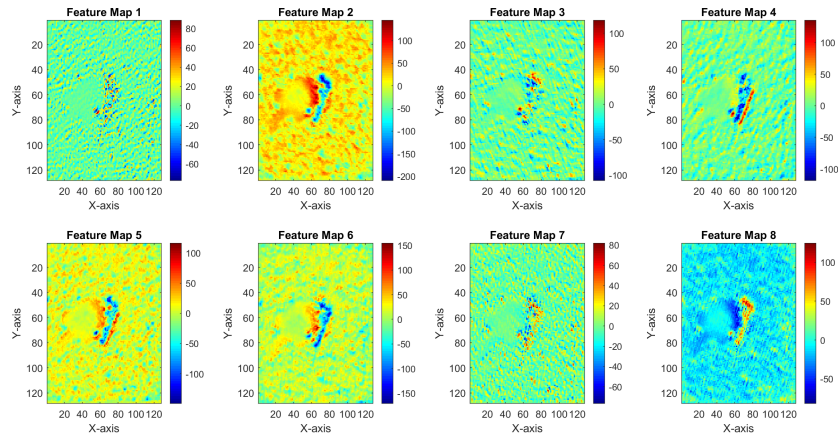


FIGURE 5. Heatmap Visualization of Convolutional Layer 1

SOC10. This comprehensive FLOPs calculation highlights the overall computational complexity of the network and underscores the importance of optimizing the model for efficient training and deployment.

V. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we present the findings derived from our proposed model, evaluating its effectiveness using both test data and new data. We conducted a comprehensive evaluation of the model's performance utilizing various metrics, encompassing accuracy, precision, and recall. Furthermore, we offer insights into the model's performance by showcasing confusion matrices derived from the experimental outcomes.

Additionally, we compared the performance of our suggested model with that of state-of-the-art methods, offering valuable insights into its relative efficacy and positioning within the research landscape. To ensure fairness and simplicity in our comparisons, we included only state-of-the-art methods that were tested on the same dataset. Notably, precision and recall metrics were often overlooked in previous studies. It's important to note that data regarding the total number of parameters and storage capacity of the models were not available for some studies. Therefore, comparisons were made only with those that provided this information. As a result, our analysis primarily focuses on comparing the overall accuracy achieved by each method.

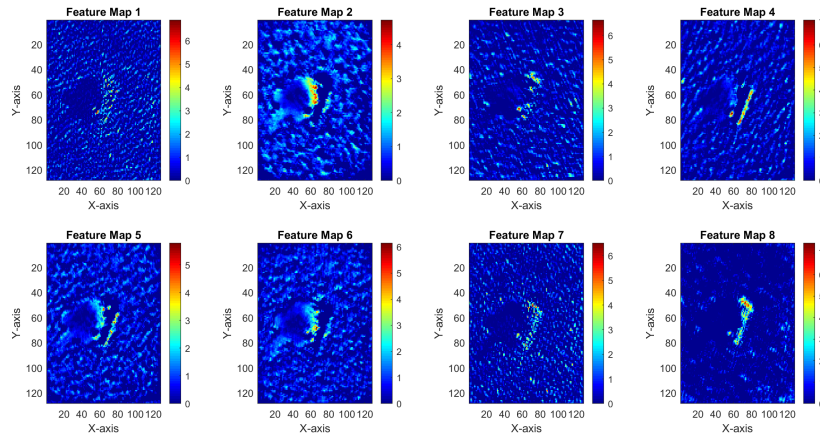


FIGURE 6. Heatmap Visualization of ReLU Layer 1

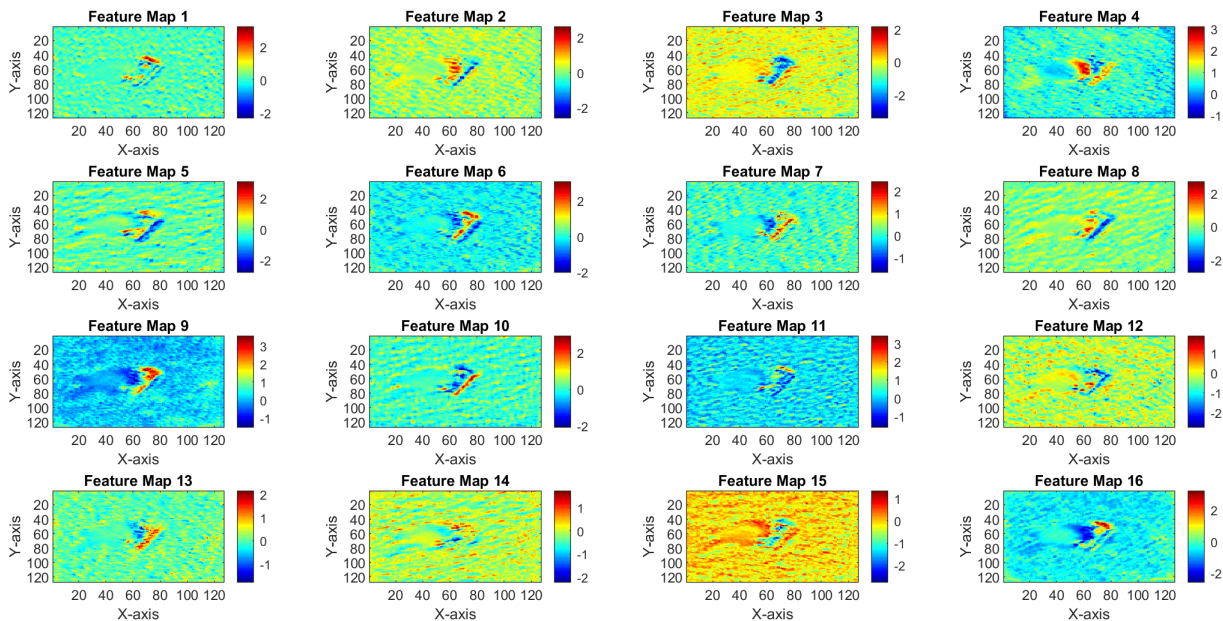


FIGURE 7. Heatmap Visualization of Convolutional Layer 2

A. PERFORMANCE ANALYSIS METRICS

The evaluation metrics utilized to assess our model are briefly summarized in this section. In our experiments, we employed accuracy, recall, and precision as evaluation metrics, with their respective definitions provided as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \quad (11)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (12)$$

and

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (13)$$

In the context of testing samples, TP, TN, FP, and FN denote sets of instances representing true positive, true negative, false positive, and false negative outcomes, respectively. TN

and TP indicate the number of correctly predicted instances, while FN and FP indicate the number of incorrectly predicted instances [31].

B. EXPERIMENTAL RESULTS ON SOC-3 DATASET

To assess the effectiveness of the LSARNet model in identifying particular targets, we utilize a confusion matrix. As depicted in Figure 9, the comprehensive confusion matrix resulting from testing the LSARNet with the SOC-3 dataset is provided. It can be observed that the LSARNet demonstrates exceptional performance, achieving 100% precision and recall for BTR70, and 99.5% precision and recall for both BMP2 and T72. The overall accuracy stands at 99.7%.

Table 10 presents a comparative analysis of accuracy, parameters, and storage requirements for various methods applied to SAR target recognition under the SOC-3 dataset.

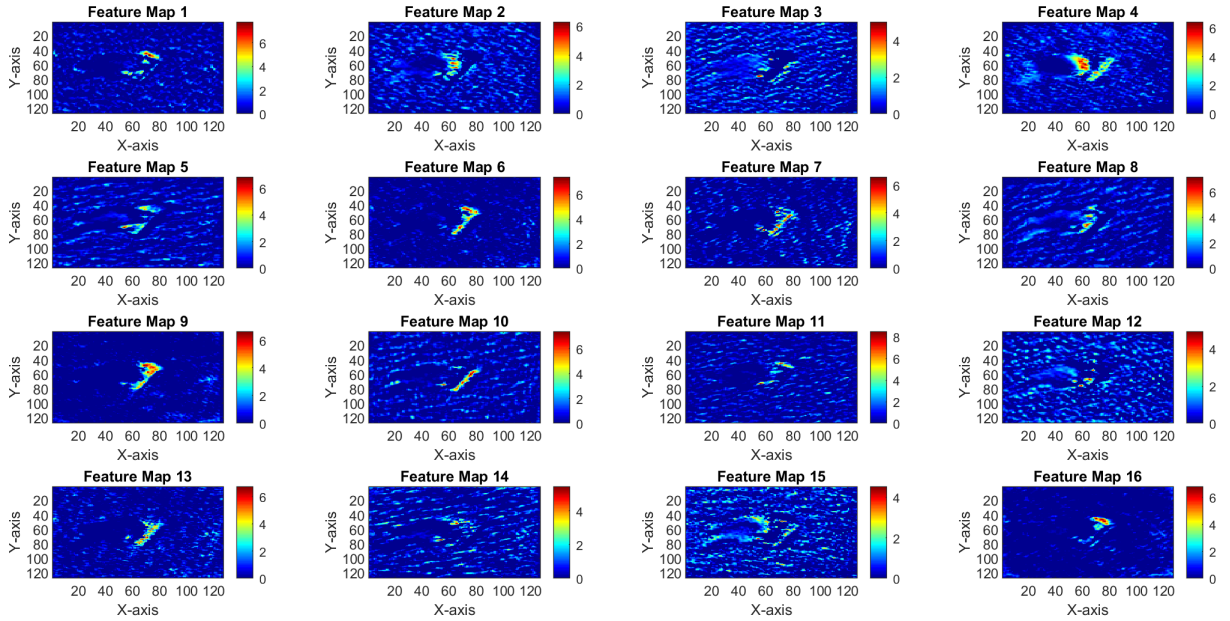


FIGURE 8. Heatmap Visualization of ReLU Layer 2

	BTR70	BMP2	T72	Precision
BTR70	196 33.4%	0 0.0%	0 0.0%	100% 0.0%
BMP2	0 0.0%	194 33.0%	1 0.2%	99.5% 0.5%
T72	0 0.0%	1 0.2%	195 33.2%	99.5% 0.5%
Recall	100% 0.0%	99.5% 0.5%	99.5% 0.5%	99.7% 0.3%
	True Label			Predict Label

FIGURE 9. Results under SOC-3

The results demonstrate the performance of our proposed approach compared to existing methodologies. Our proposed method achieves a significantly higher accuracy of 99.7%, outperforming previous techniques such as CVLWNet, CNN+SVM, and SARNet. Notably, our method also exhibits superior efficiency in terms of model parameters and storage requirements, with only 767,019 parameters and 2.9 MB storage space. These findings highlight the effectiveness and efficiency of our proposed approach in addressing the challenges of SAR target recognition, particularly in resource-constrained environments.

TABLE 10. Comparison with other Models under SOC-3 Dataset

Model	Parameters	Storage (MB)	Accuracy (%)
SARNet [2]	7,070,931	28.3	98.3
CVLWNet [32]	-	-	99.6
CNN+SVM [33]	-	-	99.5
FaDDL [34]	-	-	99.4
LSARNet (Our)	767,019	2.9	99.7

C. EXPERIMENTAL RESULTS ON SOC-10 DATASET

To offer a clear visual representation of the precision, recall, and accuracy rates across all classes, we utilize a confusion matrix. Illustrated in Figure 10, the comprehensive confusion matrix resulting from testing the LSARNet with the SOC-10 dataset is presented. Analysis of the confusion matrix reveals that LSARNet attains a precision of 99.6% for BRDM2 and recall rates of 99.5% for both BTR70 and T72. The overall accuracy stands at 96.0%.

Table 11 provides a comparative analysis of accuracy, parameters, and storage requirements for various methods applied to SAR target recognition under the SOC-10 dataset. Our proposed approach is evaluated alongside existing methodologies to assess its performance in this context. While our method achieves an accuracy of 96.0%, it demonstrates competitive results compared to previous techniques. Additionally, our proposed method exhibits efficiency in terms of model parameters and storage requirements, with 2,545,138 parameters and 9.0 MB storage space. These findings underscore the effectiveness of our approach in addressing the challenges of SAR target recognition across a broader range of classes, positioning it as a viable solution for practical deployment in real-world scenarios.

	2S1	BMP2	BRDM2	BTR60	BTR70	D7	T62	T72	ZIL131	ZSU234	Precision
2S1	239 9.9%	0 0.0%	2 0.1%	0 0.0%	1 0.0%	0 0.0%	2 0.1%	0 0.0%	1 0.0%	0 0.0%	97.6% 2.4%
BMP2	0 0.0%	188 7.8%	3 0.1%	0 0.0%	0 0.0%	0 0.0%	1 0.0%	0 0.0%	0 0.0%	0 0.0%	97.9% 2.1%
BRDM2	1 0.0%	0 0.0%	255 10.5%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	99.6% 0.4%
BTR60	3 0.1%	0 0.0%	4 0.2%	190 7.8%	0 0.0%	0 0.0%	0 0.0%	1 0.0%	0 0.0%	0 0.0%	96.0% 4.0%
BTR70	1 0.0%	2 0.1%	1 0.0%	1 0.0%	195 8.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	97.5% 2.5%
D7	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	271 11.2%	0 0.0%	0 0.0%	0 0.0%	3 0.1%	98.9% 1.1%
T62	4 0.2%	0 0.0%	0 0.0%	2 0.1%	0 0.0%	0 0.0%	254 10.5%	0 0.0%	0 0.0%	0 0.0%	97.7% 2.3%
T72	10 0.4%	5 0.2%	2 0.1%	0 0.0%	0 0.0%	0 0.0%	5 0.2%	195 8.0%	1 0.0%	0 0.0%	89.4% 10.6%
ZIL131	16 0.7%	0 0.0%	6 0.2%	0 0.0%	0 0.0%	3 0.1%	9 0.4%	0 0.0%	271 11.2%	1 0.0%	88.6% 11.4%
ZSU234	0 0.0%	0 0.0%	1 0.0%	2 0.1%	0 0.0%	0 0.0%	2 0.1%	0 0.0%	1 0.0%	270 11.1%	97.8% 2.2%
Recall	87.2% 12.8%	96.4% 3.6%	93.1% 6.9%	97.4% 2.6%	99.5% 0.5%	98.9% 1.1%	93.0% 7.0%	99.5% 0.5%	98.9% 1.1%	98.5% 1.5%	96.0% 4.0%
True Label											

FIGURE 10. Results under SOC-10

TABLE 11. Comparison with other Models under SOC-10 Dataset

Model	Parameters	Storage (MB)	Accuracy (%)
CNN [6]	-	24.2	99.5
CVLWNet [32]	-	18.7	98.2
FaDDL [34]	-	-	93.3
LSARNet (Our)	2,545,138	9.0	96.0

D. COMPARISON WITH STATE-OF-THE-ART MODELS

We conducted a thorough comparison of our model with several state-of-the-art counterparts outlined in the literature. This included an analysis of models featured in studies by [2], [35], and [32]. In Table 12, we present an in-depth comparison of our proposed method with these state-of-the-art models for SAR target recognition, covering both SOC-3 and SOC-10 datasets. The table showcases the accuracy, parameters, storage requirements, and GFLOPs of various methodologies applied to the task. For SOC-3 dataset, our proposed method achieves a remarkable accuracy of 99.7%, outperforming previous techniques such as RepLKNet-31B, DenseNet121, VGGNet, A-ConvNet, and CV-CNN. Notably, our method demonstrates superior efficiency in terms of model parameters, storage, and computational complexity, with only 767,019 parameters, 2.9 MB of storage space, and 0.16 GFLOPs.

For the SOC-10 class, our proposed method also exhibits competitive performance, achieving an accuracy of 96.0%. While our accuracy is slightly lower compared to certain state-of-the-art models such as RepLKNet-31B and A-ConvNet, our method maintains efficiency with 2,545,138 parameters, 9.0 MB of storage space, and 0.17 GFLOPs. These findings underscore the effectiveness and efficiency of

our proposed approach in addressing SAR target recognition tasks across different classes, positioning it as a promising solution for practical deployment in real-world scenarios.

VI. FUTURE WORK

Building upon the findings of this study, several avenues for future research can be explored to enhance the applicability and performance of the proposed lightweight CNN model for SAR image analysis. First, testing the reliability of the model when implemented in Internet of Things (IoT) systems is crucial. Evaluating its performance in resource-constrained environments typical of IoT applications will provide practical insights into its robustness and effectiveness in real-world scenarios. Second, conducting an ablation study to analyze the variation in input image sizes would be beneficial. By testing the model with diverse input conditions, we can assess its adaptability and resilience, ultimately enhancing its generalization capabilities. Finally, addressing the performance gap observed on the SOC10 dataset is imperative. Investigating enhancement modules or alternative techniques to improve accuracy in challenging datasets will strengthen the contribution of this work and expand its applicability. These efforts will further establish the proposed model as a reliable solution for SAR image classification, particularly in resource-limited applications.

VII. CONCLUSION

The classification of SAR targets has undergone significant refinement with the advent of machine learning techniques. However, a prevailing challenge persists: many existing methods in the literature demand substantial computational resources and memory allocation. Notably, CNN models commonly utilized in SAR target classification often comprise numerous layers, resulting in large model sizes that can hinder practical deployment. In response to this challenge, our research proposes a novel solution: a lightweight CNN architecture characterized by just two convolutional layers. Our Lightweight CNN (LSARNet) framework is specifically designed to strike a delicate balance between model size reduction and the preservation of high classification accuracy. Through rigorous experimentation and evaluation on the MSTAR dataset, our findings underscore the effectiveness of LSARNet in achieving superior recognition accuracy compared to existing CNN-based methods. Moreover, what sets our approach apart is its inherent simplicity and lightweight nature, achieved by prioritizing the extraction of salient features over the arbitrary deepening of neural network architectures. Looking ahead, our research lays the groundwork for future endeavors to further refine and optimize the proposed LSARNet architecture. By focusing on enhancing its performance across a spectrum of tasks and datasets, we aim to unlock its full potential in real-world applications. Additionally, exploration of novel optimization techniques holds promise for even greater reductions in storage requirements without compromising the model's accuracy. Ultimately, our work contributes meaningfully to the ongoing advancement

TABLE 12. Comparison with state-of-the-art Models

Dataset	Model	Parameters	Storage (MB)	Accuracy (%)	FLOPs (G)
SOC3	RepLKNet-31B [36]	-	-	99.6	-
	DenseNet121 [37]	-	-	99.3	-
	VGGNet [38]	-	-	99.3	-
	A-ConvNet [26]	-	-	99.6	-
	CV-CNN [39]	-	-	98.24	-
	Alex-Net [40]	20,157,123	80.6	93.6	-
	Le-Net [41]	4,252,573	17	97.3	-
	ResNet [42]	859,299	3.5	97.7	-
	LSARNet (Our)	767,019	2.9	99.7	0.16
SOC10	RepLKNet-31B [36]	-	78.8	97.0	0.75
	ResNet18 [42]	-	42.7	96.5	0.40
	DenseNet121 [37]	-	27.1	96.4	0.58
	AlexNet [43]	-	98.2	93.6	0.53
	VGGNet [38]	-	131.5	95.8	0.63
	A-ConvNet [26]	-	68.1	97.9	0.68
	CV-CNN [39]	-	73.0	92.7	0.42
	MFCNN [44]	11,500,000	-	95.5	-
	CNN-AD [45]	12,000,000	-	89.0	-
	VersNet [46]	5,900,000	-	99.0	-
	LSARNet (Our)	2,545,138	9.0	96.0	0.17

of lightweight deep learning models, offering new avenues for efficient deployment in practical scenarios and reaffirming the transformative potential of machine learning in SAR target classification.

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